

The bias-variance tradeoff

Readings for today

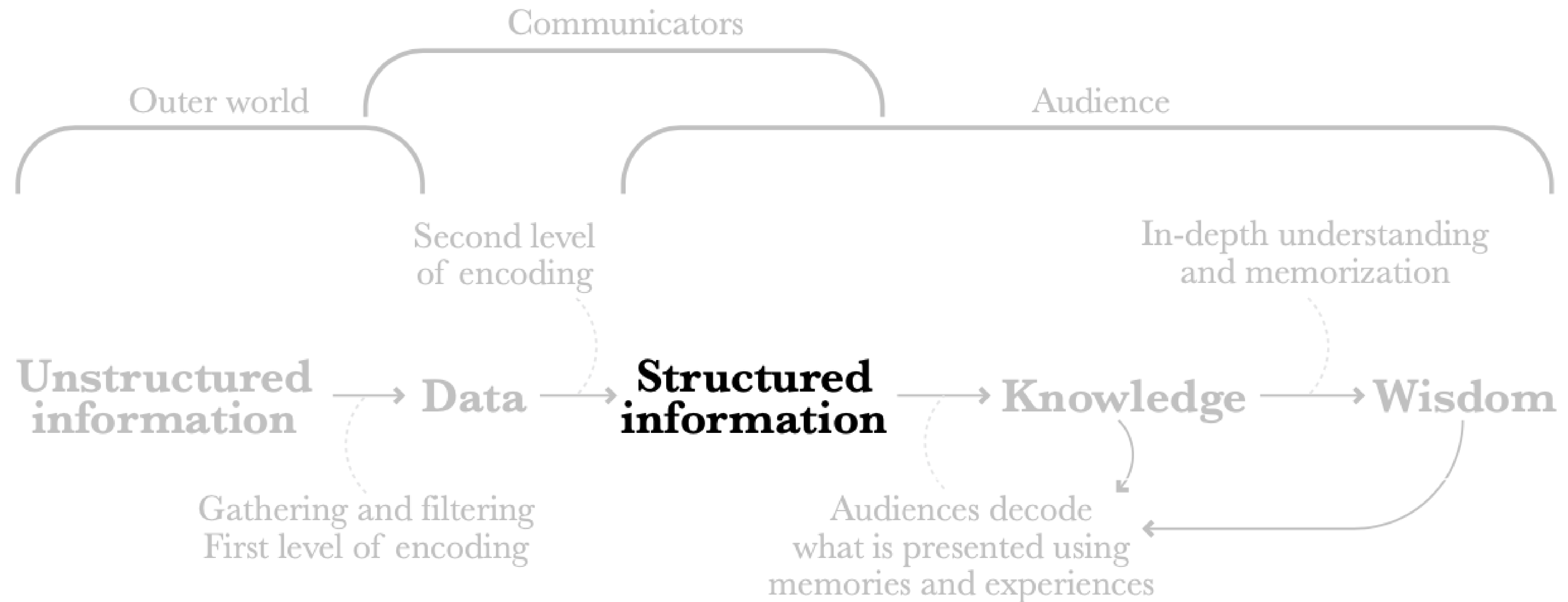
- Chapter 1: Introduction. James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). An introduction to statistical learning: with applications in R (Vol. 6). New York: Springer.
- Chapter 2: Statistical learning. James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). An introduction to statistical learning: with applications in R (Vol. 6). New York: Springer.

Topics

1. Fundamental form of statistical models
2. Bias-variance tradeoff

Fundamental form of statistical models

From data to wisdom



Theory → Hypothesis → Statistical Model

Fundamental form of a theory

$$\underset{\substack{\uparrow \\ \text{capacity}}}{C} \leftarrow \underset{\substack{\uparrow \\ \text{input}}}{f(I)} = \underset{\substack{\uparrow \\ \text{output}}}{O}$$

theory

(van Rooij & Baggio 2020)

Fundamental form of a statistic

$$\underset{\substack{\uparrow \\ \text{output}}}{Y} = \underset{\substack{\uparrow \\ \text{input}}}{f(X)}$$

relational function

f

The form of a statistical test, f , is a quantitative description of a specific hypothesis being evaluated (whether or not a p-value is calculated)

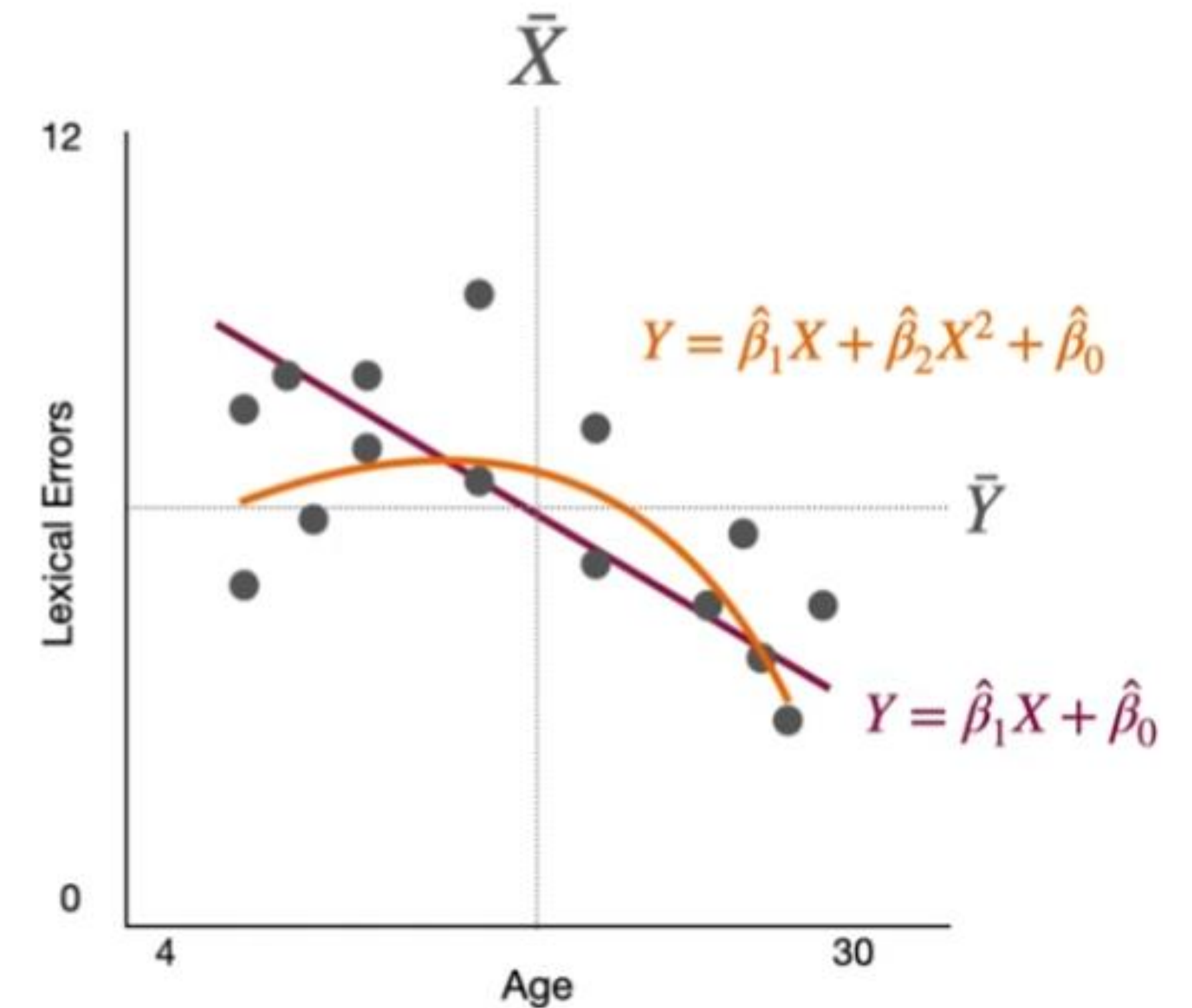
Fundamental form of a statistical model

output
↓
 Y
↑
 $\begin{pmatrix} y_1 \\ \dots \\ y_n \end{pmatrix}$

input
↓
 $f(X)$
↑
 $\begin{pmatrix} x_{1,1} & \dots & x_{1,p} \\ \dots & & \dots \\ x_{n,1} & \dots & x_{n,p} \end{pmatrix}$

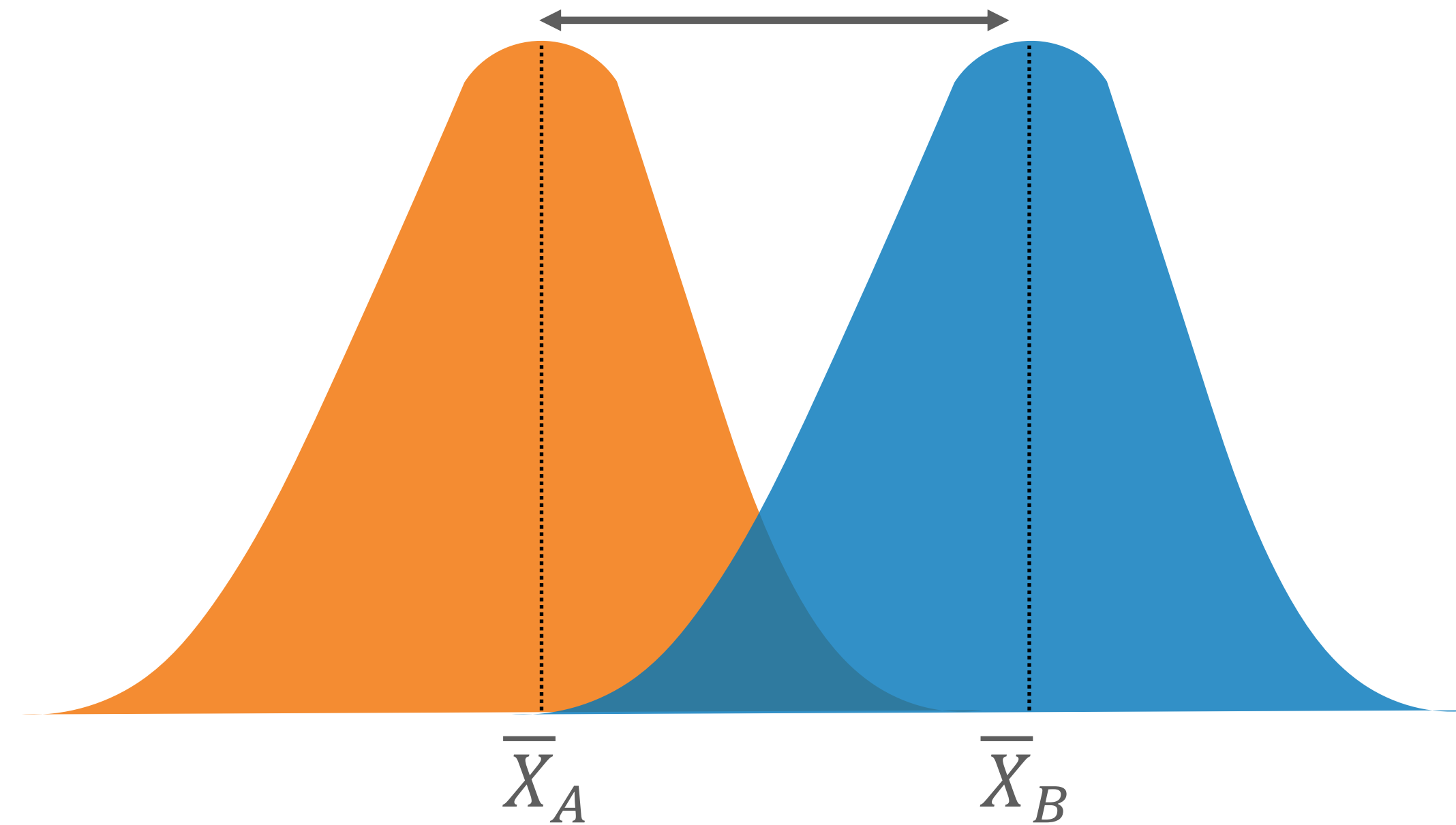
“noise”
↓
 ϵ

$$Y = f(X) + \epsilon$$



Example: t-test

T-Test: $t = \frac{\bar{X}_A - \bar{X}_B}{\sigma_{A,B}} = \frac{E(X_A) - E(X_B)}{\sigma_{A,B}}$



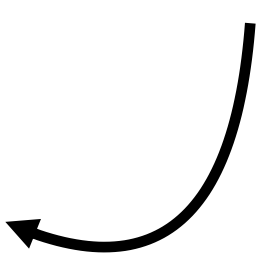
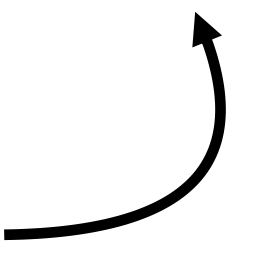
Example: t-test

T-Test: $t = \frac{\bar{X}_A - \bar{X}_B}{\sigma_{A,B}} = \frac{E(X_A) - E(X_B)}{\sigma_{A,B}}$

$$\begin{pmatrix} 100 \\ 232 \\ \dots \\ 452 \end{pmatrix} = Y = f(X) = f\left(\begin{pmatrix} 1 \\ 0 \\ \dots \\ 1 \end{pmatrix}\right) = \beta_1 X + \beta_0$$

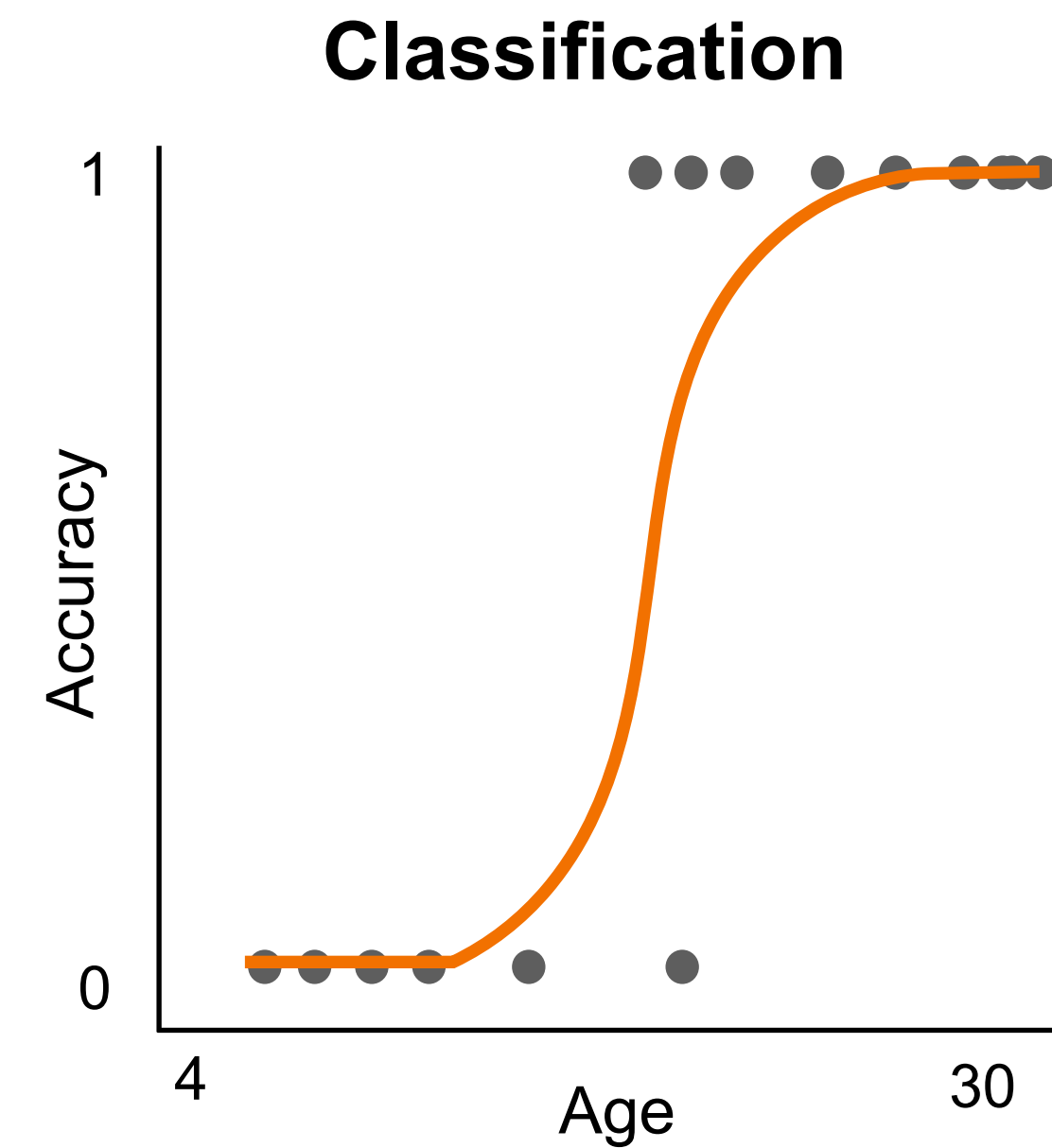
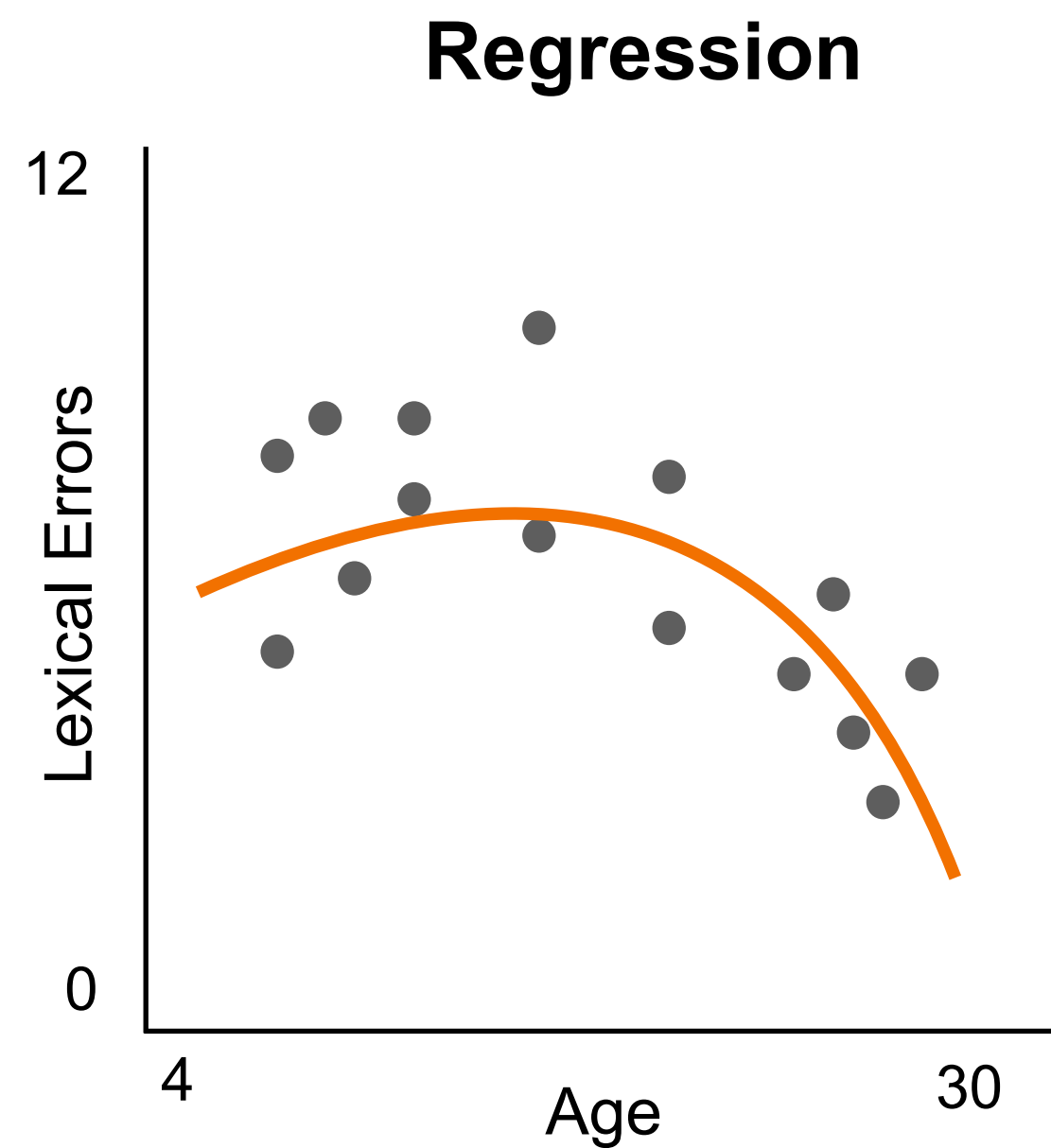
If group A, otherwise 0

Goals of learning f

1. Prediction: predict a future observation in Y from X . 
2. Inference: understand how changes in X associate with changes in Y . 

Classes of models

	Y	
	Quantitative	Categorical ('Qualitative')
Regression	✓	
Classification		✓



Forms of f

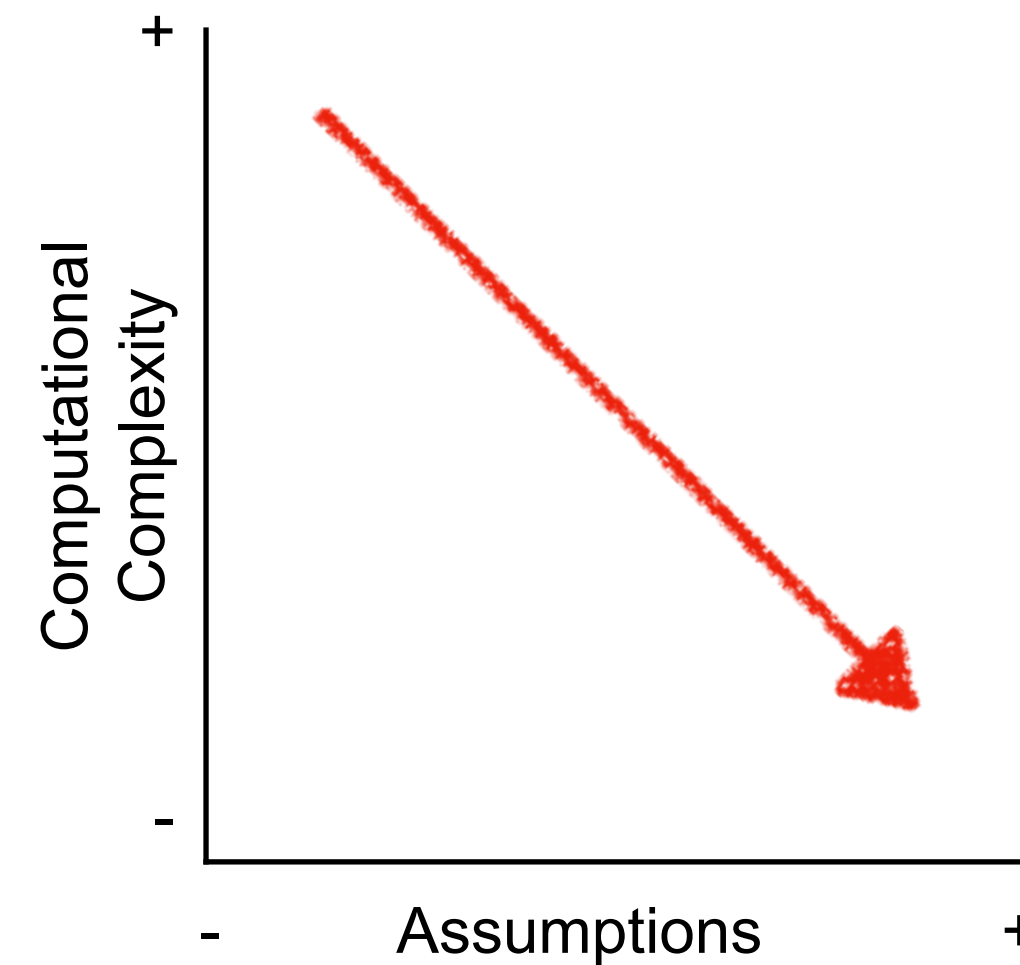
1. Parametric: assume a functional form of f that can be described by a small number of parameters.
2. Nonparametric: estimate f as close to the data as possible.

Linear regression

$$f(y_i|x_i, \beta_1, \beta_0, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-\sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_i))^2}{2\sigma^2}}$$

kNN regression

$$f(y_i) = \frac{1}{K} \sum_{i \in K} y_i$$



Bias-variance tradeoff

Dimensions

Dimensions of a model

n: number of observations (i.e., rows)

p: number of features/independent variables (i.e., columns)

Dimensions of a model: n x p

As $n \rightarrow p$, dimensionality increases

$$\begin{pmatrix} x_{1,1} \\ \dots \\ x_{n,1} \end{pmatrix} \rightarrow \begin{pmatrix} x_{1,1} & x_{1,2} & \dots & x_{1,15} \\ \dots & \dots & \dots & \dots \\ x_{n,1} & x_{n,2} & \dots & x_{n,15} \end{pmatrix}$$

Learning f

1. Supervised: find a model that minimizes a loss function on X and Y .

$$\sum_{i=1}^n (y_i - \hat{y}_i)^2$$

2. Unsupervised: find f independent of Y .

$$\operatorname{argmin}_S = \sum_{i=1}^k \sum_{x \in S_i} \|x - \mu_i\|^2$$

Types of error

Error:

$$\begin{aligned} E[(Y - \hat{Y})^2] &= E[(Y - \hat{f}(X))^2] \\ &= E[(f(X) + \epsilon - \hat{f}(X))^2] \\ &= \underbrace{[f(X) - \hat{f}(X)]^2}_{\text{Reducible}} + \underbrace{\text{Var}(\epsilon)}_{\text{Irreducible}} \end{aligned}$$

$Y = f(X) + \epsilon$

1. **Reducible Error**: Error that can be explained by $\hat{f}$.
2. **Irreducible Error**: Error that you can have no control over (i.e., noise).

Flexibility & Generalizability

$$E[(Y - \hat{f}(X))^2] = Var(\hat{f}(X)) + [Bias(\hat{f}(X))]^2 + Var(\epsilon)$$

↑
expected
error

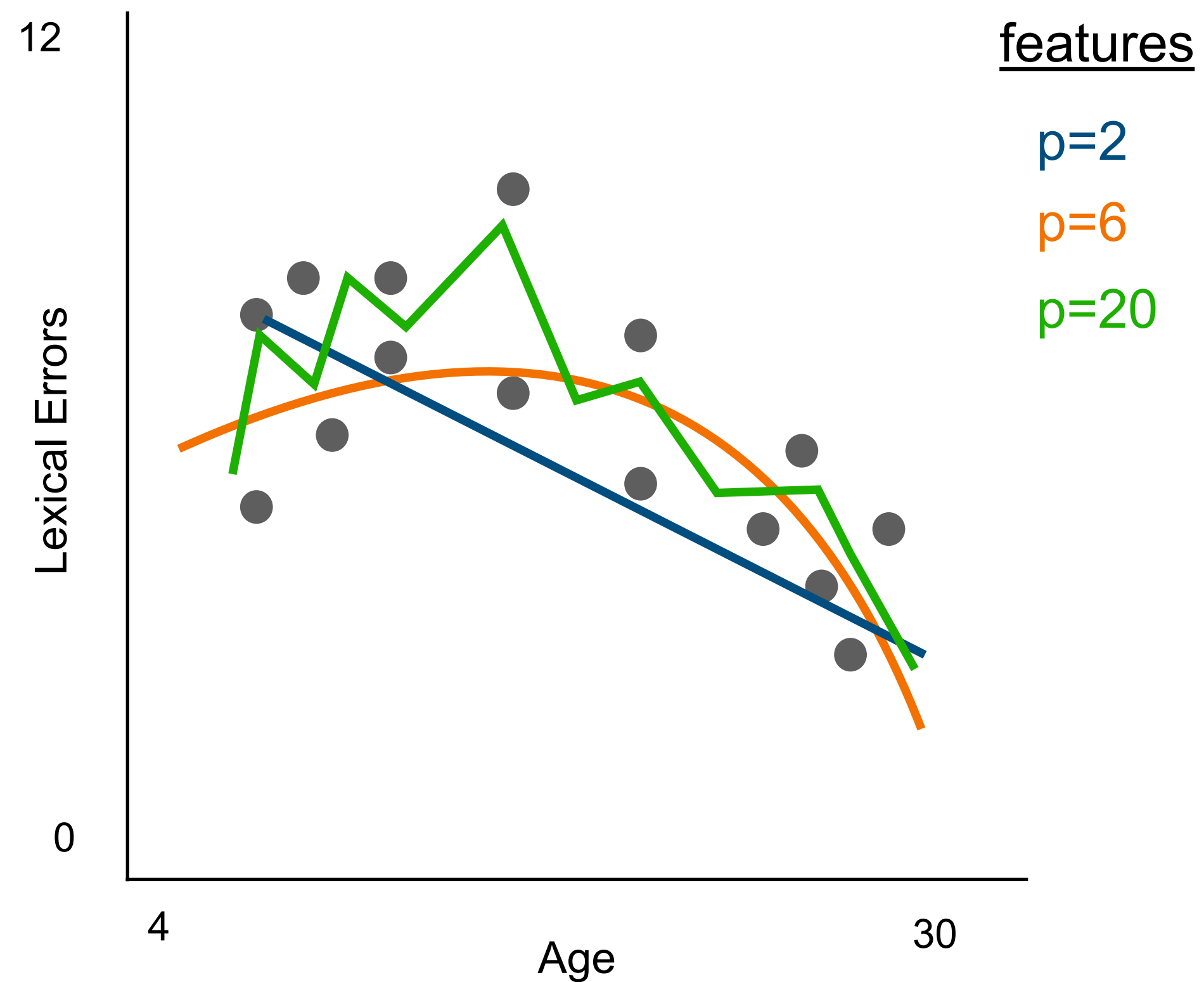
↑
how $\hat{f}$ changes with
different X & Y

↑
how well $\hat{f}$ generalizes
to a new sample of Y

↑
the unexplainable

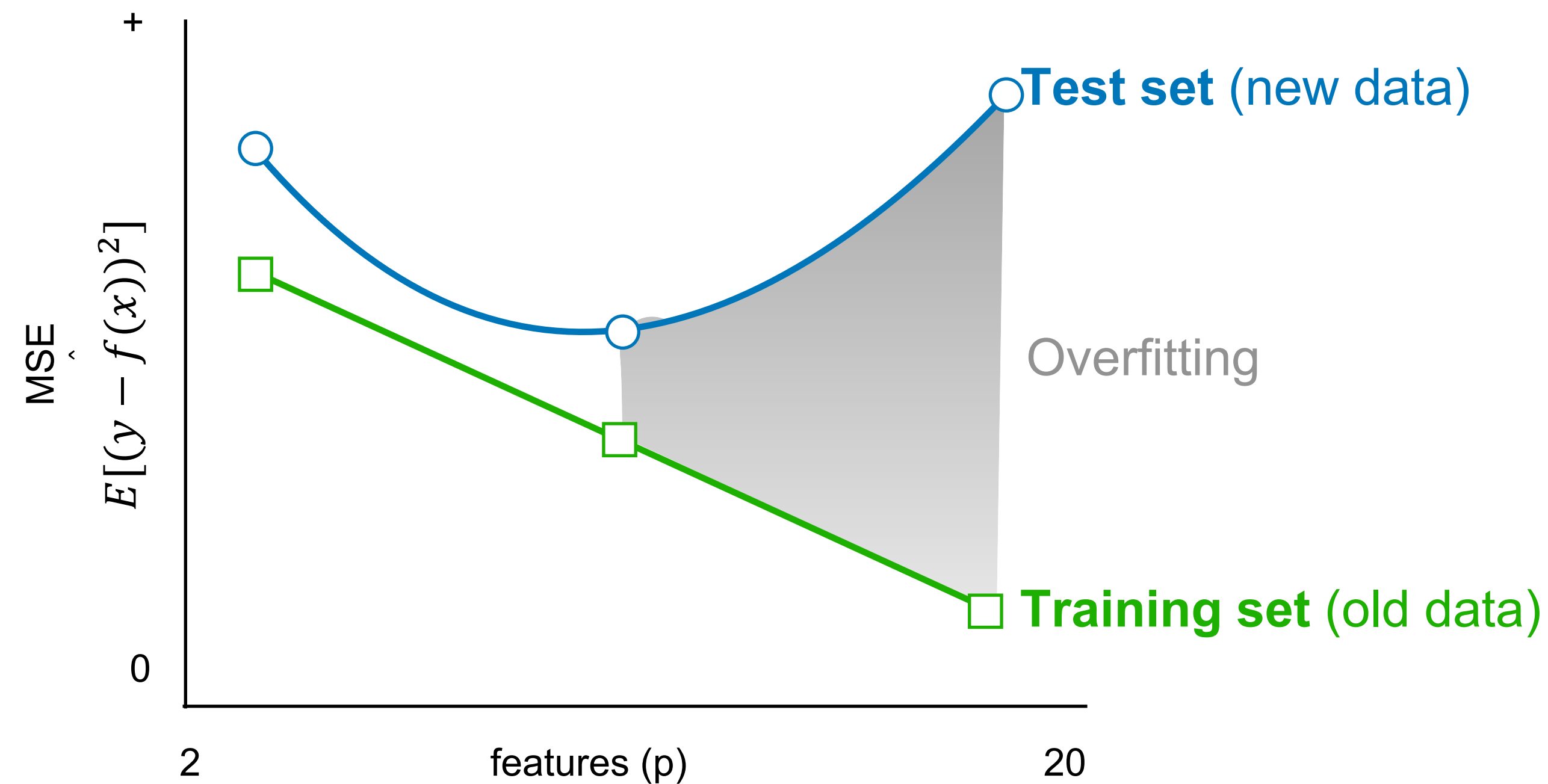
Bias-variance tradeoff

Bias-variance tradeoff



$\uparrow$ *flexibility* = $\downarrow$ *bias*

Bias-variance tradeoff



Goal: Find the right complexity that balances the flexibility (variance) of a trained model with its bias.

Take home message

- The fundamental form of all relationships in statistics is $Y = f(X) + \epsilon$.
- The best models account for reducible error in a way that explains the most variance (flexibility) while retaining the ability to generalize to new data (bias).